Opportunities to build Food System resilience and sustainability in the Elgon Sub-region and African-Secondary City: Mbale City

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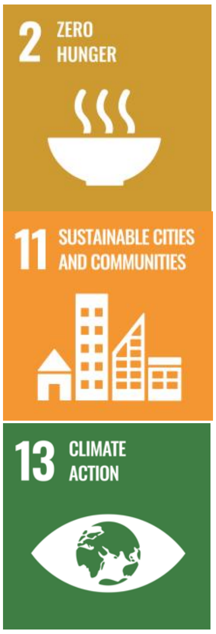
Soil, Food, Sustainability

Index

Addressed Sustainable Development Goals (SDGs)



Code availability: all codes used to compute the 3 Figures displayed below are reproducible at this repository link: [fener95/GEE-bugisu-project: A collection of scripts used for my takehome exam on Sustainable Food Systems (github.com)]

Statement of research questions:
1.Is the parish development model (**PDM**) able to increase the food system sustainability and resilience capacity in the Elgon sub-Region?
2.Are there challenges and opportunities to fill gaps?
3. Is it possible to measure and track the progression towards sustainability and resilience in the subregion and within this development context?

1. Background: Urban food system in African secondary cities and mountain ecosystem in the peri-urban-rural areas

Under the shared common goal of leaving no-one behind in the present and future Food System, here is presented a tentative approach to explore and mitigate two major challenges in one case of the African Food System: increasing pressure of small cities urbanization on the Food System and the re-known higher exposure to shocks and crisis of smallholding farmers in remote locations of an urban-peri-urban and rural area, in particular due to climate-related shock and stresses as along the slopes of the Mt Elgon, whose farmers are at high risk due to flash floodings reported seasonally (during wet seasons), and furthermore unsustainable agricultural practices until recent years, added to spread lacking of water and soil management (SWM) systems(1),(2). Putting at risk the whole ecosystem revolving around the Mt Elgon and can decrease the overall resilience capacity of the Food System in this sub-region of Uganda.
At the same time, urbanization is characterizing the area, with increasing bi-directional demanding of resources from and to the growing small city and towns along this developing urban-rural catchment area (1), (7).

2. Study area: brief of topography and climate

Home to both Sebei and Bugia cultures, the name of this subregion comes from the extinct volcano, Mt. Elgon, and approximately one million people reside in this hilly area. In **figure1** (**code available** at: GEE-bugisu-project/Fig1_BugisuProject.ipynb at main · fener95/GEE-bugisu-project (github.com)), we can appreciate the distribution of altitude zones in the subregion, starting from about 1000m in the Elgon-sub-region western-side, which connect Elgon to the wider Eastern-Region of Uganda and, in east-direction, up to altitudes over 4000m where we find the highest peak of Mt Elgon, at 4,321 meters(3).
Climate is humid-sub-tropical, with average temperatures around 23 °C and average annual rainfall around 1500m of precipitation per year (4). Besides human disturbance on soil and forested area that have had a severe impact on the ecosystem, also the topographical and climate characteristics makes this area particularly susceptible to heavy run-off, flooding and landslides events; with reported extensive damages and life losses (5a), (5b).
Generally, four different broad vegetation communities can be observed. Mixed montane forest can be found up to elevations of 2500 m, bamboo and canopy montane forest can be found from 2400 to 3000 m, and moorland can be found above 3500 m. However, natural vegetation is heavily influenced by human activities. Because of the pressure from the rapidly increasing population, vegetation is damaged by intense agriculture and grazing activities, especially in the area below 2200 m, indeed, great part of the landscape is occupied by crop fields (Bamutaze,

Yazidhi & Pilesjö, Petter, 2014). The Elgon sub-region is here defined as the total extension of the 8 districts around Mount Elgon, namely: Mbale, Sironko, Bududa, Kween, Kapchorwa, Manafwa, Bukwo and Bulambuli (Uganda Bureau of Statistics:09_2023UGANDA_DEMOGRAPHIC_AND_HEALTH_SURVEY_(UDHS)_2022_KEY_FINDINGS.pdf (ubos.org)).

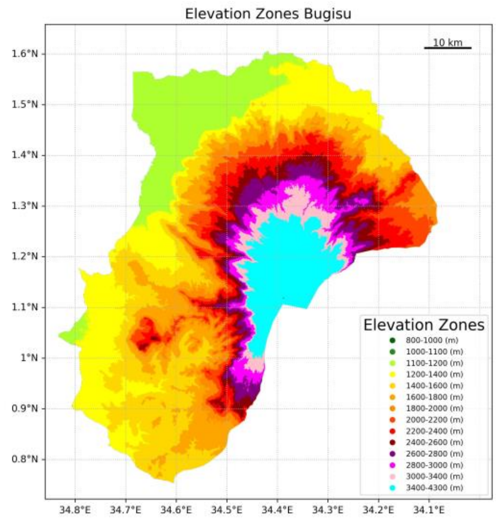


Figure.1 'Elevation zones in Bugisu. Code: from-data-source-to-map: **LINK.**

3.Current Development National Policy in Uganda

"From subsistence to a money-economy".
Launched in 2022, the Parish development Model (PDM), following the National Development Plan III (NDPIII), the NRM 2021-2026 manifesto and Sustainable Development Goals-no poverty (SDG 1) and zero hunger (SDG 2), and in line with the Local Govenment Act, 1997 which establishes local government authority and recognize parishes as the smaller economic and political unit of the Ugandan Democracy, with the final goal of decentralizing decision at the community level, empowering democracy in the Nation.

This development plan is a mixture of private and public interventions to rise the socio-economic conditions of most smallholder farmers which represent, according to the last Uganda National Household Survey (UNHS, 2019/20), 39% of Ugandan smallholder-farmers who produce for subsistence with little or nothing left to produce added value and income increase.

The main points of the PDM are establishment of parish-level Organizations (here, PO) of producers in the primary sector, individuate leadership positions in the parishes organizations and boost community engagement and coordination and collaboration in the development policy actions, within the boundaries of their parish. Finally, most relevant is the commitment of all the PO's members to adhere to the PDM SACCO (savings and credit cooperatives organizations); yet, here (12), are defined some shortcomings of the PDM: lack of clear plan of how to use savings and credits for development investments by the PO; missing definition of a common vision towards the current, expected and unexpected threats from a variable climate, current and future economic crisis and other shocks and stresses which may undermine the results and effort of the

PO and local government to realize a sustainable development, with special regards for the Food System, that along with being cross-sectorial, is also specifically addressed in the policy context.

Coordination of CSO-led project within the PDM

The PDM is also an effort by the government to minimize the need to rely on external sources of fundings and microcredits from international financing institutes and NGOs, y means of a general improvement of the socio-economic conditions of Ugandan smaller farmers

Indeed, as reported in the case of the pilots eba-projects, as a jointly effort of both international organizations and the Ministry of Water and Environment of Uganda, carried in the higher lands settlements of Elgon(14) and providing real interventions in the area, with reported outputs and technical guidance by formulating and providing a manual (15) on eba in the mountain ecosystem of Elgon, at the use of both farmers and mostly for the agriculture extension services officers. Thus, as also stated by researchers from the Makerere University (12), a PDM should also integrate the possibility to engage with CSO (NGO, Intergovernmental bodies, and others).

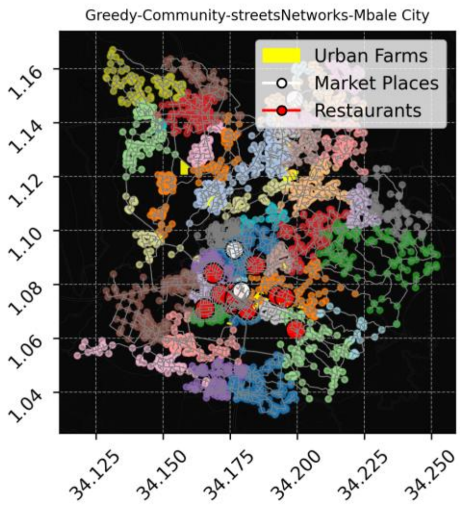
4.Challenges and opportunities along the urban-rural-continuum

According to the last 2024-census (10 years after the last one launched and published in 2013-14) based on preliminary results (National-Population-and-Housing-Census-2024-Dissemination-of-Preliminary-Results, "Uganda Bureau of Statistics" (UBOS), 2024"), Uganda is a noticeably young Nation, with 50.5% below 18 y.o., and a total working age population (14-64 y.o.) of 55.6%. This clearly pose the question if current policy of development is meeting the needs and challenges faced by the Ugandan rural and urban youth first, secondly the demanding of an increasing urban area as in the agglomeration of Mbale City, create a great challenge in the planning of development, trying to avoid trends that points to slums degradation and (silent) urban poverty and food insecurity in the lower income and less connected households.(8) Already, Mugagga, F. et al. (2013) addressed the challenges of implementing urban and peri-urban agriculture and forestry systems (UPAF) along the urban continuum, which is mostly affected by old regulations on land use planning, furthermore the urban surface in expansion, should match the demand of land for agricultural purpose in the context of a UPAF. Thus, at first fundamental infrastructure development is advised, along with revised urban multifunctional land planning and favoring exchange of knowledge.

4.1 Urbanization and Mbale City: Food system challenges and strategies

Mbale City represents a special place among the secondary cities of Uganda (where we will consider as primary city only the Kampala Capital City and the classification of urban agglomerations is intended as and based on building surface size and population density at the 'grid(also said cell or pixel)-level' as in the classification of the Urban Rural Catchment Area by Cattaneo et al., 2021)(9)), being this 'small-city' the largest urban agglomeration in the Elgon sub region and with Jinja, the largest Eastern secondary city of Uganda (2024-census-preliminary-results-UBOS, 2024).

Mbale City has an extension of 156 Km^2, 289.000 total population within the municipality and a population density of 1.863 people per Km^2, the highest in the Elgon sub-region where the average population density, Mbale City included, is 549 (population/ squared Km).
According to the city development plan (2020), as reported in “The State of the city-food system report – Mbale"(11), key points of commitment are land use planning, urban water resources management and waste disposal; moreover, ensuring safe and available food market access, stressing mostly post-harvest practices along with the current commitment of unions and organizations involved, for example, in the Parish organizations, aggregating production and resources that may open for financial access, and lastly but probably most impacting will be the training of an increasing youth involved in urban-peri-urban and rural agricultural activities and other sectors.
Regarding projects and actions directed to the food market access, the ‘Good Food for cities’ program has been operating since 2022 in 5 years long project (17), addressing specifically the hygienic practices and spaces organization in the Mbale central Food Market (in **figure.2(code available at:** GEE-bugisu-project/Fig2_BugisuProject.ipynb at main · fener95/GEE-bugisu-project (github.com); is the extension of the streets network of Mbale City, retrieved from Openstreetmap with osmnx(OSMnx 1.9.4 documentation) library and processed with networkX to classify communities area based on streets edges and modularity of the network(source code here: greedy_modularity_communities — NetworkX 3.3 documentation), and then mapped with Matplotlib(18), the network is then overlaid over urban farm, restaurant and marketplaces as available in the Openstreetmap repository); indeed , the foundational step of the project is to build multi-stakeholders participation in the development and actuation of policies.
Finally, one of the agricultural resources available in the nearby peri-urban area are rice paddy fields. Rice cultivation has been pointed as the cause of wetlands encroachment in Mbale district area, due to an uncontrolled spread of paddy fields within wetlands area, moreover, this cultivation alone accounts for an estimated 48% of green house gas emission from crops(19). On the other hand, rice, due to some cultivars resilient capacity to tolerate floodings (20), makes it an logical choice for those farmers working in the lower catchments of the region, affected by seasonal floodings. Adding to this point, recent and ongoing projects, led by CSO(21), are delivering technical assistance to foster this production in the area providing support by the ‘SUSTAINABLE RICE PLATFORM’ and through a manual on practices(22) of rice cultivation with respect for sustainability indicators and preserving surrounding wetlands which provide ecosystem services in face of drought spells and floodings events.

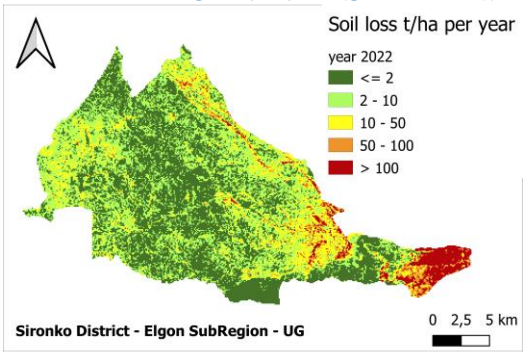


4.2 Sustainable solutions: Ecosystem-based-Adaptation

Nature based solutions (**NBS**) proved to be cost-effective in reducing risks and preserving the ecosystem services (6), indeed, cost-effectiveness is a key-enabling driver for the adoption of those kind of climate-smart solutions in developing nations. Moreover, **ecosystem based adaptation** (eba), which is a typology of NBS, that besides being cheaper than grey-infrastructure development, it is also more flexible than engineers-led solutions for adaptation to a variable climate.

They are usually 'flexible' and easy to re-module when needed, and when they are effectively taught to the recipients of the community that makes use of them, they have been proved to be effective (6).

Finally, they are a more 'flexible' and thus adaptable instrument to face shock and stress from climate-related events (see, respectively, flood shocks and drought stress to both crop systems and the community; as well as human-related threats like soil erosion due to excessive soil disturbance and deforestation, here in **Figure.3,** is figured the classification of Soil loss classed (t/ha per year in 2022), computed via RUSLE equation, as further explained in paragraph 5, point 1. Code available at: [GEE-bugisu-project/Fig3_BugisuProject.ipynb at main · fener95/GEE-bugisu-project (github.com))]



Challenges of implementing eba-measures

We assume, for the scope of this report so, that nature-based solutions, like ecosystem based-adaptation (eba), are capable of building sustainability and resilience capacity to the whole Food system, addressing at the same time the three dimensions: social, environmental and economic sphere. Then, as stated in (16), once a project supporting eba has been initiated, numerous challenges are posed to its continuation in time. The major risk of eba failure locally, is due to inefficient coordination and communication between the parties of the project, namely local government, civil society organizations (CSO) like NGO and intergovernmental bodies and the community people. Plus, many shortcomings are in the project-end reporting and follow-up, especially the case of **missing outcomes** in results, along with the outputs. The last, **outputs**, are more related to the specific intervention taken and its direct results (for example, the intervention is wetland restoration and the direct outputs can be area of wetland restored at the end of the project; whilst **outcomes** relate to the 'issue' itself, i.e. water shortages due to prolonged drought spells, in this case the outcome can be measuring 'the quantity of water saved from drought'.

5. Addressing Sustainability & Resiliency of Food system from Parishes Level

Supposing that, without opportunity of phenomena measurements along with data collections, is inherently impossible to build a total resilience capacity, added to the necessity for a food system to be sustainable. Hence, having clear and simple parameters to measure is paramount to tackle variability of anything. Here a flow of actions, summarized in 3 simple points (Point 0 and Point1 and Point 2) that can deliver opportunities of data collections, leveraging earth observations, remote sensing and GIS technologies to improve the Uganda national report on the SDG 2.4.1 ‘area of agricultural land under productive and sustainable practices’, subtheme soil degradation, by computation of the RUSLE factor(10) and suggestions:

Point 0, is recurrent and comprises three main parts:

a. PO are established, and their interconnections at the parish level are mapped, with also geolocations of the farmers plots (for example with an ID assigned to each along with their coordinates and giving notes on the crop type selected from the pool of options available for the PDM model. The PO themselves should individuate and rank their own challenges in terms of production, sustainability, and then report it. Farmer ID with confirmed geolocations from last meeting, crop type planted, and practices selected from the given pool are always updated and so reported.
b. During the pre-scheduled meetings, the PO gives updates on their ongoing practices to support a sustainable, resilient and profitable production. Following one or more of the proposed practical strategies by the committee.
c. Changes are operated, in particular eba strategies allow for a fast remodulation of the interventions as stated in previous paragraph on eba pilot projects.

Point1

a. Figure 3 is a random P-factor (practices on crop area) soil loss case-scenario, in which ‘Practices are implemented in each pixel-cell as a conditional operation attributing one value for P from a list of constant values, related to the degree of slope in the same pixel-cell. This fictitious scenario may become real if point 0 is fulfilled and data analysis is externalized to both local authorities and CSO, through mediation of the PO leaders.
b. Stratification of the characteristics of each PO, by topography, by typology of suggested, adopted sustainable practices. Then, assessing the connections between and within POs, mapping the stakeholders in this parish-based food system, stressing the grade of **influence** of each **actor** in the **agri-value chain** (who brings more value in the production? Which commodity?), in the **community engagement capability** (which actor or body possess connecting capabilities?), and **financing access** (who brings the greatest potential to attract micro-credits and national and foreign investment opportunities?)

Point2

The final aim is to guide the local food system within a methodological approach that maximize the **full resilience capacity** in all its 5 components, for instance:

a. **Prevention**: after assessing major threats and needs, exploit the CSO technical assistance to deliver hybrid grey-green solutions, with a strong foundation in ecosystem based-approaches to prevent expected shocks; Rice plantation in the lower lands around Mbale City, in respect of wetlands restoration and sustainable agronomic practices to reduce rice-paddies methane emissions (sources: NO-202308-ST-EN-SRP-Standard-for-Rice-Cultivation-V2.2.pdf (sustainablerice.org) and (source: Global programmes | Sustainable rice (rikolto.org).

b. **Anticipation**: existence and access to **Early Warning Systems (EWS),** empowering farmers with **Climate information Tools** (**CIT**). CilT needs to be supported by infrastructure development; currently as in the most case of smallholders farmers in LIC (low income countries), mobile phones and sms re the most viable ICT solutions to communicate rapidly with and among farmers (source:Designing effective digital-based delivery of climate information for smallholder farmers: a mini meta-analysis on drivers and barriers: Climate Policy: Vol 0, No 0 - Get Access (tandfonline.com)). Moreover, Forecast-based finance is particularly relevant in the case of damaged crops and farm facilities by natural events like floods which affects reportedly the downstream catchments in the area;
c. **Absorption**: supportive communities and social protection programs; for instance, after stratification of smallholders, the PO is able to identify those farmers at higher risk of being damaged by flash floodings or prolonged drought spells, in this case one portion of the provision of PDM-SACCO funds could be redirected to support those farmers at risk, so both anticipating and absorbing the impact; clearly resources from the fund are limited and are, by policy, prioritize to enhance general value of the agrifood production, monetizing this economy. However, it is also true that making transparent and public data on the agricultural cover and practices and geolocation of the plots, may enable both research public institutions and privates of any size, to better analyze and forecast-demanded support for affected producers.
d. Adaptation and Transformation: having continuous data collection and analysis, permits to set a 'continuous improvement' mentality which thanks to the basic foundations on eba practices, can enable fast change and recovery from a current situation. Finally, transformation is obtained by 'changing things' considering incoming new insights. In practice, this may result in changing supply chains and distribution channels, so discussing how actors are interconnected among them, below a secondary-city region food system lenss.
e. Key commodity in the area is the bugisu coffee cultivar, arabica and robusta coffee in an mix-banana-coffee cropping system widely spread in the higher altitudes (mostly between 1600-2200m) are predominately governed by the bugisu and Sebei farmers unions(source:Elgon-Investment-Profile.pdf (ugandainvest.go.ug)), and such an established agri-value chain may be fertile soil for application of more technology intense tools, like **blockchain** (source:Blockchain technology for land registry management in developing countries — ResearchOnline (gcu.ac.uk), which can also address the omnipresent issue of land rights and entitlement that undermined many development projects in the area.

6.Discussion

The example provided in this report on the condition of Food system of the urban rural catchment area of Mbale in the Elgon region; depicted that ecosystem-based adaptations options revealed feasibility in terms of cost-benefit, and furthermore, enhanced the whole ecosystem resilience. The discussion here is on the urge to create a common vision for the whole Elgon area, integrating all the policy related to a complex Food Systems, for example, ensuring that the new PDM model , a policy directed to address smallholders farmers with the aim to lift them from poverty via creation of parishes-level agri value chian, and increase community joined interest towards the food system, also thanks to the creation of POs and their links to the local government. It is suggested, to include a participation of CSO (ngo and other provisioners of funds, technical skills and project) who have proved to increase sustainable and resilient approaches, as in the case of the Ecosystem based adaptation programs exposed here, carried along the periurban-rural area.

Moreover, and mostly, according to this report, it is need to embrace a secondary-city region food systems , in an effort to build on existing value adding projects for urban resilience with policy 'touching' all the urban rural continuum and specific CSO-supported projects (via bilateral financial agreements) to integrate resources and effort to create a whole resilient food system under sustainable and productive prerogative for the whole agri-food sector.
In this order, Parishes communities, locals and district governmental bodies should individuate the vary and continuous variable needs along Urban-Rural linkages, and act so to direct funds in the appropriate actions, sharing a common vision building on the Elgon residents, identity promoting occasion of social gathering and recognition of virtuous work along the whole food supply chain.

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